

Design of an Automotive Interior Motion Detection System using PIR and GSM Integration

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Abstract—Conventional vehicular alarm systems have historically focused on detecting external threats, yet they neglect the safety of vehicle occupants who may inadvertently become entrapped inside a parked cabin. Incidents involving children or pets locked in vehicles, as well as concealed intruders, underscore the critical need for interior motion detection solutions. This paper presents a compact, cost-effective automotive interior motion detection system that integrates a Passive Infrared (PIR) sensor, a PIC16F877A microcontroller, and a GSM communication module. The system activates upon the simultaneous satisfaction of predetermined parking conditions: engine off, handbrake engaged, and doors locked, and subsequently monitors the vehicle cabin for any thermal motion signatures. Upon detection, the GSM module transmits an SMS alert to the vehicle owner in near real-time, independent of proximity constraints. The architecture is designed for straightforward integration with existing vehicle platforms, requiring no significant modification to embedded software or electrical systems. Empirical validation via a working prototype demonstrates reliable detection and timely notification delivery, establishing the proposed system as a viable enhancement to contemporary vehicle safety mechanisms.

Keywords- Motion detection, PIR, CPD, GSM modules, PIC microcontroller

I. INTRODUCTION

In the past, automotive security systems were designed to counter exterior perimeter threats by using localized alerts to prevent car theft. However, this perimeter-centric paradigm ignores the crucial safety of the inside cabin, especially when it comes to the presence of covert attackers or the unintentional imprisonment of sensitive people, including children and pets. Regulatory agencies, such as the European New Car Assessment Programme (Euro NCAP), have encouraged the incorporation of Child Presence Detection (CPD) technologies in contemporary automobiles because rapid thermal escalation within a sealed, parked vehicle can cause life-threatening hyperthermia in a matter of minutes. To address these dual challenges of safety and security, this report evaluates an event-driven interior motion detection architecture comprising a Passive Infrared (PIR) sensor, a deterministic RISC microcontroller, and GSM-based telemetry. Rather than proposing a novel implementation, this document critically analyzes the operational methodology, hardware justification, and environmental limitations of this specific tri-component baseline. By evaluating the system's reliance on thermal transients and digital-threshold processing, this report contextualizes its

efficacy and power efficiency against the stringent operational constraints of modern automotive environments.

To overcome the intrinsic drawbacks of early occupancy detectors, active sensing modalities have received a significant amount of attention in recent years. The “static target” vulnerability is the main cause of this change. High-frequency active emission can effectively pass through physical barriers (such as blankets or chairs) to map the interior environment, according to studies assessing Frequency Modulated Continuous Wave (FMCW) radar [1]. In a similar vein, studies using commodity Wi-Fi Channel State Information (CSI) have demonstrated that detection accuracy of over 95% may be attained through active electromagnetic monitoring [2]. Because they can identify micro-movements, such as a sleeping baby's rhythmic chest displacement, these active devices are highly valued in recent literature because they successfully address the static-target issue that besets passive sensors.

The inadvertent entrapment of children and animals in parked vehicles constitutes a significant and recurring public safety hazard. Rapid internal temperature escalation within enclosed automobile cabins can produce life-threatening hyperthermia within minutes, a phenomenon extensively documented in forensic and paediatric literature [3]–[5]. In the United States alone, approximately 882 child fatalities attributable to vehicular heatstroke were recorded between 1999 and 2022, representing an annual average of approximately 38 deaths [6].

Children under six years of age are particularly vulnerable, as developmental limitations preclude independent egress from a sealed vehicle. These statistics provided the impetus for the European New Car Assessment Program (Euro NCAP) to introduce incentive provisions in 2022 for manufacturers that incorporate Child Presence Detection (CPD) technology into new vehicle models [2].

Beyond the risk of accidental entrapment, vehicular security threats arising from concealed intruders represent a further motivation for interior monitoring capability. Instances in which individuals secrete themselves within a parked vehicle to ambush unsuspecting drivers highlight a vulnerability inherent to conventional perimeter-only alarm systems, which are inadequate for detecting intra-cabin threats after the vehicle has been secured [1]. Furthermore, unattended valuables left

inside parked vehicles in high-footfall environments such as shopping centers and multi-storey car parks remain attractive targets for opportunistic theft.

Interior motion sensing addresses these deficiencies by providing real-time surveillance of the vehicle cabin during periods of non-occupancy. Unlike proximity-limited technologies such as Bluetooth or Radio Frequency Identification (RFID), which operate within ranges typically below one metre, Global System for Mobile Communications (GSM)-based notification architectures can deliver alerts across geographically unrestricted distances via established cellular infrastructure [7], [8].

II. STRUCTURAL ARCHITECTURE & HARDWARE SELECTION

In order to realistically classify between humans and animal intruders within the enclosed, tightly constrained space of an automobile cabin, it becomes necessary to use a system architecture which relies on a highly streamlined hardware profile consisting of a passive sensor interface coupled with a dedicated edge microcontroller. The minimization in this structural design solves two major problems: it reduces the probability of firmware errors during execution as well as addresses the severe power consumption constraints on parked vehicles. Architecturally, it becomes an ideal choice to use a Passive Infrared sensor due to its micro-ampere power consumption, targeted Field of View covering $110^\circ \times 70^\circ$ via specialized Fresnel optics.

A. Monitoring PIR Sensor

Monitoring is accomplished via the pyroelectric concept, which involves the sensor detecting changes in thermal radiation caused by human movement across its field of view. The sensor measures the thermal transients induced by a moving heat source, not the actual temperature [6]. PIR sensors struggle to identify motionless human bodies. If a child falls asleep and remains fully immobile, the lack of thermal movement results in a flat signal, which could cause a missed detection [6], [9]. PIR sensors were a major component of earlier Child Presence Detection systems, however these systems frequently had limited coverage inside the intricate geometry of a car interior and had a high false alarm rate because of temperature fluctuations in the surrounding environment [10].

The principal design objective is to provide a standardized, platform-agnostic solution that resolves the ‘island problem’ associated with proprietary vehicle architectures—that is, the incompatibility of safety add-ons across different embedded software environments [11].

The monitoring system cannot be “always on” since PIR sensors are prone to false alerts from driving or external movement. State machines are used in modern in-car presence detection to decrease interference and adaptively determine the target’s state [6]. The microcontroller functions as an edge node that depends on the Body Control Module or Electronic Control Unit of the car. Until the ECU broadcasts that the car satisfies the stringent requirements for CPD monitoring, the

microcontroller stays in a dormant state. The PIR sensor does not start monitoring until after that.

B. System Architecture

In order to achieve this, the system architecture is deliberately kept minimal: a single PIR motion detector, a GSM communication module, and a PIC microcontroller constitute the complete hardware assembly. This simplicity confers dual advantages: it minimizes computational overhead, thereby reducing the probability of software errors, and it facilitates maintenance and long-term reliability.

The system operates in a strictly event-driven manner, remaining dormant until a defined set of activation conditions is concurrently satisfied. These conditions are:

- The vehicle must be stationary and in a parked state.
- The ignition must be deactivated.
- The transmission selector must be in the park (P) position.

Upon confirmation of all three conditions, and with all doors in a closed and locked state, the embedded microcontroller initiates the PIR sensor monitoring routine. Any motion detected within the cabin is immediately relayed to the GSM module, which dispatches an SMS or email notification to the registered vehicle owner [6]. A major advantage of non-invasive sensors like PIR is their simplicity. Because the microcontroller only needs to process a binary or low-bandwidth analog signal from the PIR, the system maintains a minimal power draw, which is vital to prevent draining the vehicle’s battery during long-term parking surveillance [12]. The system architecture is illustrated in Fig. 1.

C. Hardware Justification

An interior motion detection system’s adherence to the rigorous energy and computational restrictions of a parked vehicle determines its architectural efficacy. As a result, the choice of the central processing unit and the sensing modality must be supported in comparison to other technologies. While recent State of the Art developments in Child Presence Detection frequently utilize active modalities such as Frequency Modulated Continuous Wave radar or commodity Wi-Fi, these systems mandate high-frequency signal processing that draws substantial continuous power [10]. The primary mandate of the proposed system is to prevent parasitic battery drain during long-term vehicle parking. PIR sensors operate strictly in the microampere (μA) range, making them highly sustainable and energy-efficient for extended surveillance [13]. Additionally, two major obstacles to contemporary in-cabin monitoring are data compliance and passenger privacy. High-resolution occupancy data can be obtained via camera-based thermal imaging or deep-learning optical systems, but these methods have significant computational overhead and serious privacy risks. PIR technology totally avoids these privacy issues while providing a clear detection trigger because it only detects thermal transients and does not render or broadcast visual images [13], [14].

A reliable embedded edge node is needed to translate the sensor’s analog-to-digital trigger. Comparing the PIC16F877A

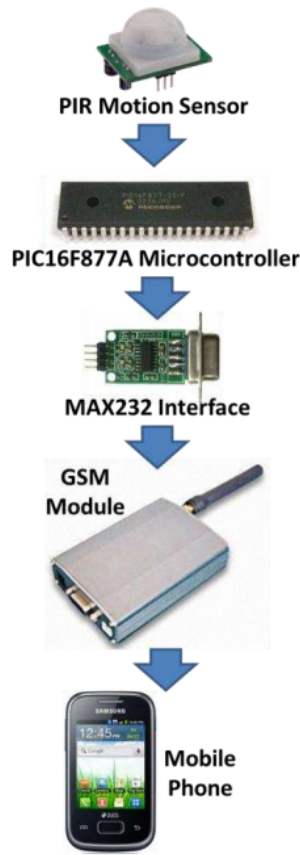


Fig. 1. Procedure of Interior Motion Detection and Notification Alert System [6].

to other widely used 8-bit RISC microcontrollers, like the ATmega328P, is very educational in order to support its deployment. The PIC16F877A is preferred for its highly predictable hardware architecture, which is recommended in industrial and automotive applications, even though the ATmega328P is frequently used in rapid-prototyping environments and has greater maximum clock speeds (up to 20 MHz) [14]. Because of its tight Harvard design and single-word instructions, the PIC16F877A ensures perfectly predictable interrupt latency, or the time it takes for the processor to react to the logic-low trigger of the PIR.

Crucially, the microcontroller must spend the great majority of its life in a dormant condition due to the vehicle's power budget. There are specific, extremely low-power sleep modes for the PIC16F877A. Until a hardware interrupt (such as the PIR signal exceeding a threshold) awakens the CPU in a few microseconds, it stops all central processing. By minimizing the needless energy expenditure associated with more complex processors or operating systems, this hardware/software co-design guarantees that a microcontroller with almost zero overhead exactly matches the computational simplicity of the PIR sensor [14].

A key advantage of GSM-based notification over alternative wireless protocols is the absence of distance constraints.

TABLE I
RFID VS. GSM MODULE COMPARISON

Aspect	RFID	GSM Module
Commands	Serial Commands	AT Commands
Distance Factor	Up to 100 m	Larger Distances
Remote Alerting	Not Good	Very Good
Weather-proof	No	Yes
Cellular Network	Not Used	Uses Cellular Network

Bluetooth and RFID technologies are effective only within distances of approximately one metre, rendering them unsuitable for remote alerting. GSM leverages the existing public cellular network, enabling notifications to be received irrespective of the owner's physical proximity to the vehicle. Additionally, GSM modules demonstrate considerable resilience to adverse meteorological conditions, maintaining reliable operation in scenarios where radio frequency-based short-range protocols may exhibit signal degradation [7]. The complete working algorithm is depicted in Fig. 2.

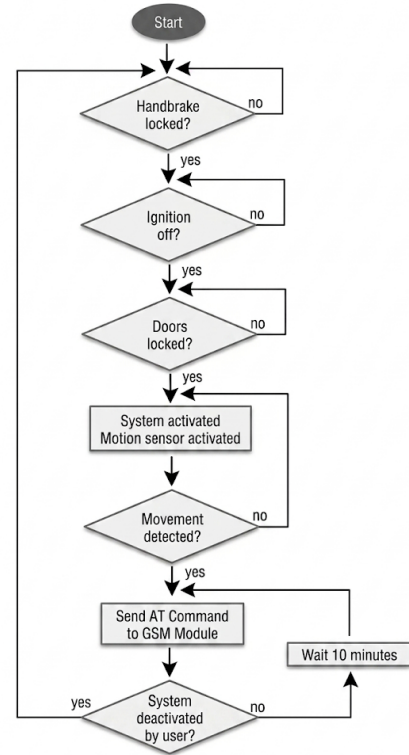


Fig. 2. The Active Working Algorithm of the System [6].

III. OPERATIONAL METHODOLOGY OF PIR SENSOR

The hardware assembly comprises three principal components: a PIR motion sensor, a PIC16F877A microcontroller, and a MOD9001D GSM/GPRS modem. Each component is selected on the basis of functional suitability, cost-effectiveness, and ease of integration [6]. A passive infrared

(PIR) detector is the central component of the interior monitoring architecture. The PIR sensor measures just the natural infrared light emitted by objects in its range of vision, operating completely passively in contrast to active sensing modalities that use radar or ultrasonic waves to map an environment [2]. Human bodies are very obvious targets for infrared detection because they usually generate thermal radiation in the 8–14 micrometer wavelength range [15]. The sensor serves as a thermal boundary monitor inside the limited space of an automobile cabin. It assesses the geographical environment for quick thermal shifts rather than absolute temperature. The sensor provides the basis for the whole security and detection pipeline by converting the physical movement of a heat source into a quantifiable electrical anomaly through the use of a highly engineered crystalline core [15].

A. PIR Sensor

The Passive Infrared sensor operates on the principle of pyroelectricity, wherein certain materials generate a transient electrical potential in response to rapid changes in temperature. The sensor element consists of two opposing pyroelectric detection areas, one with positive polarity and one with negative polarity. The sensing element is composed of a crystalline material that possesses a spontaneous electrical polarization. When infrared radiation strikes this material, it induces a rapid temperature change, which alters the crystal's atomic structure and displaces bound surface charge [15].

The resulting induced current is directly proportional to the rate of temperature change over time, expressed by the equation:

$$I = p A_s \frac{dT_s}{dt} \quad (1)$$

where I is the generated current, p is the pyroelectric coefficient of the specific crystal, A_s is the surface area of the sensing element, and $\frac{dT_s}{dt}$ is the rate of change of the sensor's temperature. This mathematical relationship dictates that a stationary heat source generates no current. To reject global ambient temperature changes, the sensor utilizes a differential configuration with two opposing elements. An alarm state is only triggered when a thermal signature moves asymmetrically across these elements [15].

In order to improve the FoV of the sensor, a Fresnel lens is used which condenses light and provides a large range of IR to the sensor. The lens is split up into multiple sections which give scattering of multiple areas [9]. The FoV minimizes dead angles where a person could otherwise go undetected by enabling a single sensor module to cast a volumetric optical net over the whole passenger compartment when mounted centrally on the interior ceiling of the vehicle. Under quiescent conditions, both elements register identical ambient infrared flux, producing a net output of zero. When a subject emitting infrared radiation (such as a human or animal) enters the sensor's field of view, the Fresnel lens concentrates the radiation asymmetrically onto the two elements, inducing a differential signal.

This differential is processed by an internal comparator, and the resulting output, typically very low in amplitude, is subsequently amplified to a level suitable for microcontroller input [15].

For vehicular interior deployment, the PIR sensor output pin is connected to the microcontroller via a 5 V pull-up resistor. The open-collector output architecture of the sensor ensures a clearly defined binary interface: the output is pulled to ground only during active motion detection, providing the microcontroller with an unambiguous logic-level signal.

The sensor is mounted on the interior ceiling of the vehicle cabin, proximate to the overhead courtesy lamp, affording a near-omnidirectional field of view across the passenger compartment [6].

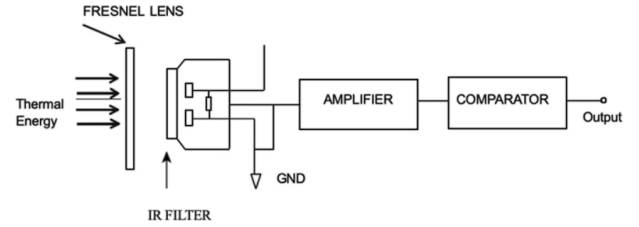


Fig. 3. A PIR Sensor [9].

B. Sensitivity

The heat difference between the target and the surrounding environment has a significant impact on the PIR sensor's sensitivity in an automotive setting [2]. The sensor measures the difference between the human body and the interior surfaces of the car rather than directly detecting the human body.

The operational sensitivity is therefore dynamic. A sharp, noticeable voltage spike occurs on a chilly day due to the high heat differential between a 37°C human and a 10°C cabin inside. On the other hand, the interior of a car can quickly reach or surpass human body temperature when it is parked in direct sunshine. The sensor's signal-to-noise ratio deteriorates when this thermal differential gets closer to zero, momentarily lowering the system's sensitivity to minute movements [2], [16].

C. Data Acquisition and Remote Telemetry

The architecture uses a tightly sequential data collecting pipeline to convert the sensor's raw physical reaction into an actionable alarm. The PIR sensor generates a binary logic-low signal upon detecting a differential heat event. This hardware interrupt is captured by the central microcontroller, which serves as the edge-processing node. The microcontroller sends the processed detection event to a telemetry module since the system runs in a setting where local auditory alarms could go unheard. The warning is sent over the cellular network by the microcontroller after being packaged into a small data payload. This telemetry architecture offloads demanding communication duties from the low-power sensing

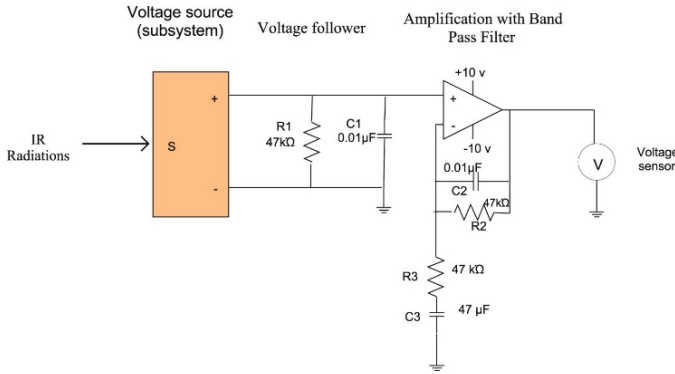


Fig. 4. Mathematical model of Sensitivity analysis of PIR Sensor [15].

node while guaranteeing that the vehicle owner receives real-time, proximity-independent messages [2].

PIC16F877A, manufactured by Microchip Technology and based on Harvard architecture, is preferred as the central processing unit of the system. The device is widely adopted in embedded systems and automotive electronic design owing to its ease of programmability, on-chip flash memory (8,000 program words), and support for In-System Programming, which facilitates iterative firmware updates without physical removal from the circuit [17].

The microcontroller operates at a nominal supply voltage of 5 V DC, regulated by an LM7805 linear voltage regulator, obviating the need for a custom power-conditioning circuit.

TABLE II
SPECIFICATIONS OF THE PIC16F877A MICROCONTROLLER

Specification	Technology Used	Comment
Architecture	Harvard	Improves execution speed
Memory	Flash Memory	Supports 8,000 program words
Programming	In-System Programming (ISP)	Allows firmware updates without physical removal from circuit
Voltage Supply	LM7805 Regulator	Eliminates need for custom power-conditioning circuit

The MOD9001D GSM/GPRS modem provides the wireless communication interface between the microcontroller and the vehicle owner's mobile device. The hardware includes a phone module, a voltage level changer, a SIM card slot and a small computer chip. The small computer chip controls the sending and receiving of messages for the phone module [18]. The modem operates within the 900 MHz to 1800 MHz frequency bands and requires a regulated 5 V DC supply. Communication between the PIC16F877A and the modem is mediated by a MAX232 voltage-level converter, which interfaces the microcontroller's TTL-level serial port with the RS-232 interface of the modem via a DB9 connector.

TABLE III
SPECIFICATIONS OF THE GSM MODULE

Specification	Technology Used	Comment
Wireless Comm.	MOD9001D GSM/GPRS modem	Provides wireless interface between MCU and owner's mobile device
Voltage Supply	5 V DC Supply	Prevents voltage fluctuations and reduces errors
Frequency Range	900–1800 MHz	Reliable for long-range communication
Level Converter	MAX232 + DB9	Interfaces TTL serial to RS-232 of modem

D. Zonal Detection and Directionality

The optical array of a conventional dual-element PIR sensor greatly influences the system's spatial resolution, even though it is essentially a binary presence detector. The cabin is divided into alternating active and passive optical zones by a dome-shaped Fresnel lens [15]. These imperceptible boundaries are crossed when an invader or imprisoned occupant moves. Since the output is only a sequence of voltage pulses, standard digital PIR systems are unable to determine the exact direction or distance of the intruder. But by monitoring the frequency of these pulses, sophisticated analog algorithms are able to categorize the movement's general profile and pace, differentiating between an animal's random motion and a human reaching across the cabin [6].

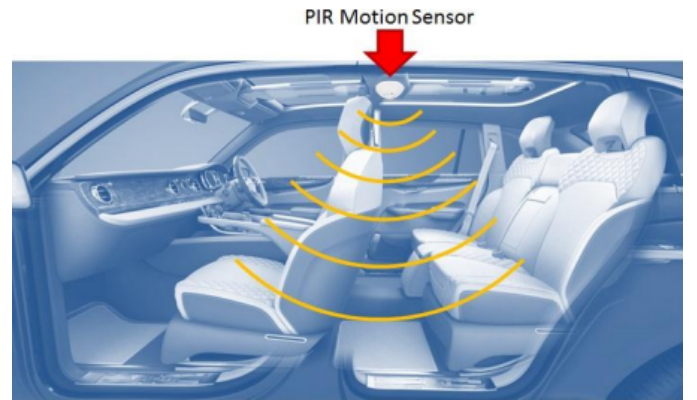


Fig. 5. PIR Sensor Placement [19].

The MOD9001D GSM/GPRS modem is an ideal choice for interior motion sensing and wireless communication, allowing owners or drivers to receive instant alerts via SMS.

This arrangement eliminates the need for ancillary signal conditioning circuitry, thereby reducing system complexity and the risk of power loss attributable to additional passive components [20]. The modem accepts standard Hayes AT commands for SMS composition and transmission, facilitating straightforward firmware integration [6].



Fig. 6. MOD9001D GSM/GPRS Modem and RS-232 cable [6].

E. Intruder Classification

The accurate distinction between human and animal presence has become a crucial prerequisite for sophisticated spatial surveillance systems in recent years. Modern signal processing techniques accomplish this distinction by separating and examining discernible morphological characteristics in the unprocessed thermal waveform. The thermal mass and surface area of the infrared source that crosses the sensor's Field of View (FoV) are closely correlated with the peak amplitude of a generated pyroelectric signal, according to empirical findings. As a result, the physiological distinctions between normal animals and humans result in clearly different amplitude profiles. Furthermore, the spatial distance between the sensor node and the thermal target fundamentally alters both the amplitude and the transient shape of the received signal. When normalized for distance, the thermal radiation emitted by an adult human subject consistently generates a substantially higher signal amplitude compared to that of a smaller animal counterpart, providing a dependable metric for algorithmic classification [9].

IV. SYSTEM EVALUATION AND COMPARATIVE ANALYSIS

A simplified architecture for automotive interior monitoring is provided by the combination of a GSM telemetry module, a microcontroller, and a PIR sensor. To validate this system, it is necessary to evaluate its operational efficacy, define its environmental limitations, and perform a comparative analysis against emerging state-of-the-art detection models.

The evaluated architecture's main advantages are its unconstrained alert routing and quick data collecting. The microcontroller can function without the latency caused by intricate analog-to-digital processing thanks to the PIR sensor's open-collector binary output [15]. The microcontroller's deterministic interrupt response guarantees that the GSM module is turned on in a matter of microseconds when a temperature transient is detected. Additionally, compared to short-range Radio Frequency (RF) protocols or isolated alerts, using GSM cellular infrastructure offers a substantial functional benefit. As long as cellular network coverage is maintained, observations of automotive IoT designs verify that GSM-based alerts effectively separate the vehicle owner from proximity restrictions, guaranteeing alert delivery over almost infinite distances [14].

However, the passive nature of the PIR sensor introduces strict operational constraints within the extreme microclimate of an automobile cabin. To produce a quantifiable rate of change, the sensor solely depends on the thermal contrast between a person and the surrounding environment. The interior temperature of a parked car can quickly rise above 37°C, which is the average human body temperature, during the hottest summer months. Thermal washout is the result of the thermal contrast becoming zero under certain particular circumstances. This can cause the system to become momentarily blind to occupant movement and significantly lower the signal-to-noise ratio of the sensor [12].

Active sensing modalities, such as Frequency Modulated Continuous Wave (FMCW) radar, have shown remarkable detection capabilities in recent assessments. The static-target and blind-spot weaknesses inherent in PIR systems are effectively resolved by radar systems, which often reach classification accuracies surpassing 95% in detecting micro-movements, such as the 4 mm chest displacement induced by newborn respiration [2]. However, because of unacceptable parasitic battery depletion, these radar architectures' continual high-frequency digital signal processing severely restricts their usability for extended parking scenarios [12]. On the other hand, recent research on sophisticated analog PIR systems improved by lightweight machine learning—like Support Vector Machines (SVM)—reports classification accuracy of up to 94% for differentiating between human and animal inhabitants [15].

The tri-component architecture assessed in this paper lacks respiratory detection and human-animal discriminating skills, in contrast to highly sophisticated models. Its deliberate architectural trade-off, however, is the omission of continuous signal processing. The system maintains an ultra-low quiescent power consumption and guarantees zero privacy threats by sacrificing classification granularity, making it a very feasible and economical baseline for extended intrusion and active-occupancy detection [12], [14].

V. CONCLUSION

This report has presented the design and prototypical validation of a PIR- and GSM-integrated interior motion detection system for automotive applications from paper [6]. The system addresses a demonstrable gap in conventional vehicle alarm architecture by providing reliable, real-time monitoring of the passenger cabin during periods of vehicle non-occupancy.

The integration of GSM communication confers a critical advantage over proximity-limited alternatives: alert delivery is independent of distance, enabling the vehicle owner to respond promptly regardless of their physical location. The system architecture is both cost-effective and modular. Its reliance on standardized hardware components and a simple conditional-activation algorithm minimizes integration complexity, making it readily adaptable to diverse vehicle platforms without necessitating modification to existing embedded software environments.

The design is also extensible: the addition of audible alarm outputs would further deter theft, while refinement of the detection threshold parameters could enhance robustness against spurious activations. Future work should address integration with modern CAN bus architectures, evaluation of alternative motion sensing modalities, and assessment of system performance across a broader range of ambient temperature and lighting conditions [21].

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